SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I C.P.Elton Pejo (192520136) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	2	2
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 24 /30 *Signature of the Faculty*

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

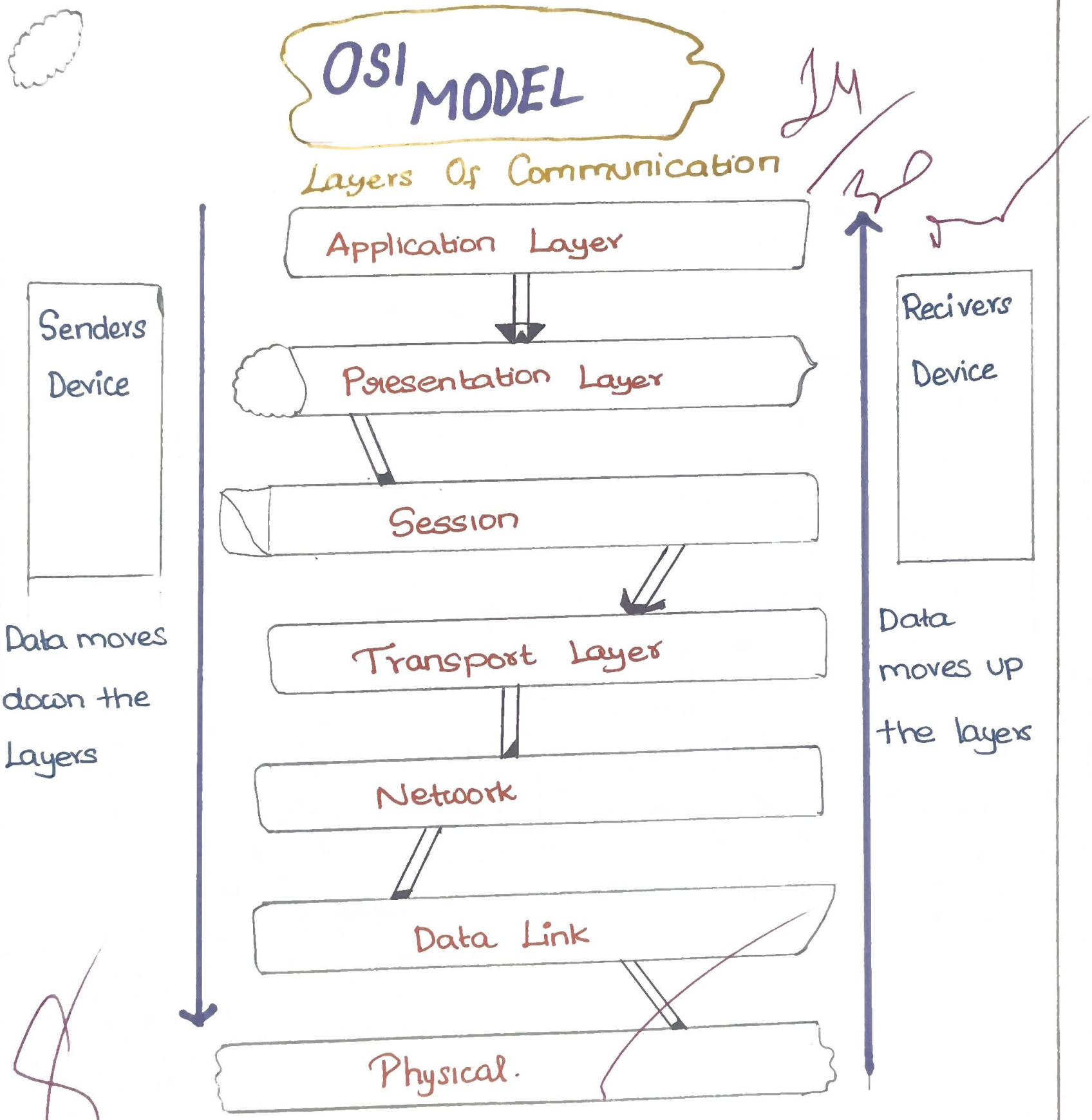
Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

COMPUTER NETWORKS COI-AT2

OSI MODEL

Layers Of Communication



Data Encapsulation

Application Data

Segment

Packet

Frame

Bits 0101 1010

Reciver reverses the process.

MAC ADDRESS VS IP ADDRESS.

Two addresses two different Jobs

MAC ADDRESS

- Data Link Layer
- Unique to each device
- Works with Local Network
- Used for frame delivery

WORK TOGETHER

Both are essential for end-end communication across networks.

IP ADRESS

- Network Layer
- Logical address
- Used For Routing
- Works across different Networks.